

Complex Transseries Reversion at Modular Cusps

Stokes corrections, flat critical values,
arbitrary ramification, and q -gamma special values

An independent study of the **ProveIt** research questions

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Abstract

Compositional inversion of a complex transseries requires a domain, a branch, and a summation prescription. We develop an explicit framework in which those data remain visible throughout reversion, and apply it to q -Pochhammer products at roots of unity. First, an elementary Mittag–Leffler calculation gives the meromorphic Borel transform of the fixed-argument cyclotomic logarithm, all its poles and residues, uniform Gevrey estimates, and its exact positive-ray Laplace sum. A nonzero residue correction disproves the unrestricted directional-equality clause of the **ProveIt** conjecture `conj:cyclotomic-resurgent-inverse`; the appropriate replacement includes a specified summation direction and Stokes corrections. The resurgence itself is established literature, which we explicitly identify.

Second, exact eta transformations reduce finite product quotients at every rational cusp to an elementary exponential–power core times a convergent series in a dual nome. When the core cancels, we prove a convergent logarithmic–Puiseux inverse, compute every coefficient, index all local inverse sheets, and give an explicit truncation bound. Three eta factors realize every ramification order and are minimal for the two eta balances. Four unnormalized Euler-product factors realize every ramification order while cancelling all three perturbative balances; four are necessary.

Third, rational q -gamma special values have exact modular completions. The half-argument completion is a theta function, whose lacunary series yields an explicit logarithmic inverse. For the original, unnormalized q -gamma value we instead prove a regular-core inverse theorem with sharp feedback scale $t_0^{-2}e^{-S/t_0}$; its leading pole coefficients are controlled by Lambert W . Exact rational calculations and high precision complex checks accompany the proofs. The results concern specified formal or analytic classes and do not assert a global inversion or summability theorem for unrestricted complex transseries.

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1 Scope, results, and relation to the repository

In this article, *series reversal* means *compositional reversion*: given a relation $y = f(t)$, determine and justify a local inverse $t = g(y)$. It does not mean reversing the order of coefficients or taking $1/f$. The central issue is that an inverse may need a different asymptotic scale from the original function. The elementary example

$$y = e^{-S/t}, \quad t = -\frac{S}{\operatorname{Log} y + 2\pi i j}, \quad j \in \mathbb{Z},$$

already exchanges an exponential scale for a logarithmic covering. If the exponential is multiplied by a power of t , Lambert W enters. If a leading correction is Q^m , a Puiseux root enters before the logarithm. If the original coefficients are divergent, a choice of Borel summation direction enters as well.

These mechanisms are often discussed separately. Here they are joined in an exact class arising from modular q -products. The resulting theorems include branch data and error bounds; they do not depend on promoting an uncontrolled formal substitution to an analytic identity.

1.1 What is proved

The main results have five distinct roles.

1. [Theorem 4.1](#) and [proposition 4.4](#) give an explicit Borel transform for the fixed-argument cyclotomic q -Pochhammer logarithm and a counterexample to one directional-equality assertion in the repository. The general resurgence mechanism is classical; the point here is to state and prove the correctly normalized specialization.
2. [Theorems 3.1](#) and [3.2](#) give two practical reversion engines. One treats an exact dominant core with a convergent exponential tail. The other treats a regular finite endpoint perturbed by $e^{-S/t}$, and identifies the sharp scale and all leading coefficient poles.
3. [Theorems 5.2](#) and [6.1](#) give the exact normal form at every rational cusp and a convergent inverse when all perturbative core terms cancel.
4. [Theorems 7.1](#) and [7.2](#) construct arbitrary finite ramification with minimal numbers of factors: three eta factors for the two eta balances, four Euler-product factors for all three raw product balances.
5. [Theorem 8.1](#) and the subsequent half-argument calculation apply the theory to rational special values of q -gamma, both before and after removal of their analytic background.

A precise negative conclusion is part of the result: a Borel ray containing no pole need not produce the same sum as another such ray. A precise positive conclusion is equally important: after fixing the summation and logarithm sheets, an inverse can be constructed with every exponential contribution retained and with a geometric remainder.

1.2 Existing results that must be credited

The source comparison is pinned to commit `6cb1d86f19ebf9f54c8d985b141b84233b6eab66`. The `ProveIt` transseries volume and companion already contain exact-core inversion, Lagrange–Bürmann formulas, exponential corrections, and positive-real q -product completions [12, 13]. The support-controlled reversion and moving-fold reports already discuss finite action grids,

grouped resonances, and analytic critical charts [14, 15]. The present argument does not claim those principles as new.

The relevant source base also extends beyond the Transseries directory. The q -Pochhammer monograph contains complex fixed-argument expansions, cyclotomic asymptotic base inverses, exact eta completions at $q = 1, -1$, ordinary finite ramification, and regular q -gamma base inversion [16]. It would therefore be incorrect to describe the repository as confined to positive real q . Its frontier chapter contains the named conjecture `conj:cyclotomic-resurgent-inverse`, whose directional-equality clause we test below.

The literature also goes substantially beyond a purely formal calculus. Van der Hoeven proves formal compositional inversion in an appropriate transseries field and develops a separate complex theory with composition-domain restrictions [1, 2]. Edgar studies composition, recursion, and convergence [3]. Marmi and Sauzin treat Borel transforms at roots-of-unity resonances for a general cohomological equation [5, Sections 4.2–4.3]. Their framework includes the Lambert series used here. Explicit quantum-dilogarithm Borel poles and Stokes jumps are developed by Garoufalidis and Kashaev [6]; more recent q -Pochhammer modular resurgence is treated by Fantini and Rella [7]. Sauzin establishes nonlinear resurgent substitution and inversion under the required analytic estimates [4].

Our contribution is an explicit development within this established framework: a correction of a specific repository assertion, exact logarithmic cusp inversion with arithmetic coefficient generation, sharp factor-count constructions, and the finite-endpoint feedback analysis for modular q -gamma corrections. These are proved below. A worldwide priority claim, or the resolution of unrestricted complex transseries inversion, would go beyond the evidence.

1.3 Notation and domain conventions

We use $\mathbb{N} = \{0, 1, 2, \dots\}$. Put

$$P(z) = \prod_{n=1}^{\infty} (1 - z^n), \quad |z| < 1.$$

This is holomorphic and nonzero in the unit disk, and its logarithm is the unique analytic logarithm with $\log P(0) = 0$. The eta function is

$$\eta(\tau) = e^{\pi i \tau / 12} P(e^{2\pi i \tau}), \quad \Im \tau > 0.$$

For an integer $n \geq 1$, let $\sigma_{-1}(n) = \sum_{d|n} d^{-1}$. A root of unity is written $\zeta = e^{2\pi i h/k}$, with $k \geq 1$ and $(h, k) = 1$; the cusp $q = 1$ is represented by $h = 0, k = 1$. We use

$$q = \zeta e^{-t}, \quad \Re t > 0, \quad \mathcal{S}_\delta = \{t : 0 < |t| < t_*, \quad |\arg t| \leq \pi/2 - \delta\}.$$

Here $0 < \delta < \pi/2$, and every asymptotic statement on $\mathcal{S}_\delta$ is uniform on the displayed closed subsector unless explicitly qualified.

The symbol Log denotes a specified branch or logarithm lift. A logarithm of a holomorphic unit equal to 1 at the origin always means its germ with value 0 there. Thus a logarithm sheet is not silently reset during continuation. For complex $q = e^{-t}$ and rational x , the definition of q^x is e^{-tx} ; this is the branch used in the q -gamma applications.

2 Complex reversion: formal data and analytic realization

2.1 Three different meanings of a transseries inverse

There are three levels of assertion. A *formal inverse* is an identity in a specified complete algebra of series, where the relevant substitutions are defined. An *asymptotic inverse* is a sectorial function with controlled remainders after truncation. An *exact convergent inverse representation* is an identity obtained by substituting small functions into a convergent analytic series. Such a representation is stronger than its asymptotic expansion, but does not supply all possible divergent or oscillatory transseries.

For example, if $U(z_1, z_2)$ is analytic near $(0, 0)$, then

$$U(e^{-\alpha/t}, e^{-\beta/t}) = \sum_{p, q \geq 0} u_{p, q} e^{-(p\alpha + q\beta)/t}$$

converges whenever both arguments lie in a suitable polydisk. It is an asymptotic transseries on a sector only when their real exponential rates decay there. If $p\alpha + q\beta$ agrees for two different pairs, their coefficients must be added before the leading term is identified. No denominator involving an action difference is needed for this addition.

Proposition 2.1 (Finite complex actions). *Let $F(u, \mathbf{z})$ be holomorphic near $(0, \mathbf{0})$, with $F(0, \mathbf{0}) = 0$ and $F_u(0, \mathbf{0}) \neq 0$. There is a unique holomorphic germ $u = U(\mathbf{z})$ with $U(\mathbf{0}) = 0$ and $F(U(\mathbf{z}), \mathbf{z}) = 0$. If $z_j(t) = e^{-\alpha_j/t}$, and a closed sector satisfies*

$$\Re(\alpha_j/t) \geq c/|t| \quad (1 \leq j \leq r)$$

for a fixed $c > 0$, then $U(\mathbf{z}(t))$ is an exact convergent sectorial transseries. Its action support is contained in $\mathbb{N}\alpha_1 + \cdots + \mathbb{N}\alpha_r$, and equal actions are grouped by finite sums at each bounded exponential height.

Proof. The holomorphic implicit-function theorem gives U and a polydisk of convergence. Shrinking the sector radius puts all $z_j(t)$ in any smaller such polydisk. Absolute convergence permits grouping. On a fixed interior ray, the real part of $\sum n_j \alpha_j/t$, divided by $1/|t|$, is at least $c \sum n_j$. Consequently only finitely many multiindices lie below a fixed height. This proves both the finite grouping assertion and the usual asymptotic interpretation of degree truncations. $\square$

The finite-action result is standard analytic infrastructure. It remains useful because it says exactly which operation is legitimate. An arbitrary countable action family would additionally require control of accumulation, coefficient summability, and substitution. An arbitrary complex direction may not meet a common decay cone at all.

2.2 Why a global complex claim fails

The size of $e^{-1/t}$ changes when $\Re(1/t)$ changes sign. The two functions

$$f_0(t) = t, \quad f_1(t) = t + e^{-1/t}$$

have the same algebraic asymptotic series on every closed right subsector, but different inverses there. Thus an algebraic expansion does not determine exponentially small inverse information. The exact normalizations in the following sections are what supply it.

A finite-point critical example is $f(t) = t^m$. Its inverse requires an m -th root; it is not an ordinary Taylor series at the critical value when $m > 1$. At a flat cusp, $e^{-1/t}$, even a Puiseux

scale alone is insufficient: a logarithm is unavoidable. For arbitrary complex exponential actions, a sector may also contain oscillatory pairs of comparable size. A total real ordering of monomials cannot simply be imported unchanged into that situation. These distinctions are consistent with the composition restrictions in the complex transseries literature [2].

2.3 A formal coefficient ring is not a summation theorem

For coefficients in a complete formal algebra, an equation

$$u = -\Phi(u, \mathbf{z}), \quad \Phi(0, \mathbf{0}) = 0,$$

with a suitably invertible linearization can be solved by coefficient recursion. In a $\mathbf{z}$ -adic algebra, every coefficient uses finitely many preceding coefficients. That fact proves formal existence and uniqueness. It does not give bounds after substituting a physical exponential scale.

For a resurgent extension, it is also insufficient that each coefficient is individually resurgent. An analytic series in a second variable must have geometric coefficient bounds in the relevant resurgent seminorms. Sauzin's nonlinear implicit theorem provides the appropriate setting [4]. Our convergent cusp theorems meet their analytic needs directly through ordinary holomorphic estimates; our Borel calculations specify the summation data rather than assuming it.

3 Two quantitative reversion engines

3.1 An exact core and all exponential coefficients

Consider a function in a phase coordinate v ,

$$F(v) = G(v) + L(e^{-\lambda v}), \quad L(z) = \sum_{n \geq 1} a_n z^n, \quad (3.1)$$

where L is holomorphic at zero. Choose a solution v_0 of $G(v_0) = w$, with $G'(v_0) \neq 0$. The solution v_0 is a *chosen core branch*; the theorem below does not claim that the core is globally one-to-one.

Define its divided difference

$$D_{v_0}(u) = \begin{cases} (G(v_0 + u) - G(v_0))/u, & u \neq 0, \\ G'(v_0), & u = 0. \end{cases}$$

It is holomorphic wherever the translated core is holomorphic. Writing $v = v_0 + u$, $\xi = e^{-\lambda v_0}$, converts the inverse equation exactly to

$$u + \frac{L(\xi e^{-\lambda u})}{D_{v_0}(u)} = 0. \quad (3.2)$$

Theorem 3.1 (Exact-core reversion). *Suppose D_{v_0} is holomorphic and nonzero on a neighborhood of $|u| \leq r$, and for some $\rho > 0$, $0 < \kappa < 1$,*

$$\left| \frac{L(\xi e^{-\lambda u})}{D_{v_0}(u)} \right| \leq \kappa r \quad (|u| \leq r, |z| \leq \rho).$$

The equation has one simple root $u = U(v_0, z)$ in $|u| < r$, holomorphic in z and any holomorphic core parameters. It has a convergent expansion

$$U(v_0, z) = \sum_{n \geq 1} U_n(v_0) z^n.$$

Put $a_n^{(j)} = [z^n]L(z)^j$. Every coefficient is

$$U_n(v_0) = \sum_{j=1}^n \frac{(-1)^j a_n^{(j)}}{j!} \frac{d^{j-1}}{du^{j-1}} \left(\frac{e^{-n\lambda u}}{D_{v_0}(u)^j} \right) \Big|_{u=0}. \quad (3.3)$$

For $\eta = |z|/\rho < 1$, the tail after degree N satisfies

$$\left| U(v_0, z) - \sum_{n=1}^N U_n(v_0) z^n \right| \leq \frac{\kappa r \eta^{N+1}}{1 - \eta}. \quad (3.4)$$

Proof. On $|u| = r$, the second term of [equation \(3.2\)](#) is strictly smaller than u . Rouché's theorem counts exactly one zero, with multiplicity. It is therefore simple. The holomorphic implicit-function theorem gives its local analytic dependence; uniqueness glues the local functions. At a root, $|U| \leq \kappa r$.

Introduce a scalar s in the equation $u = -sL(ze^{-\lambda u})/D_{v_0}(u)$. Lagrange–Bürmann gives

$$u = \sum_{j \geq 1} \frac{(-s)^j}{j!} \partial_u^{j-1} \left(L(ze^{-\lambda u})^j D_{v_0}(u)^{-j} \right) \Big|_{u=0}.$$

For a fixed coefficient of z^n , only $j \leq n$ contributes, so coefficient comparison is finite and is valid formally before any evaluation of s . Since

$$[z^n]L(ze^{-\lambda u})^j = a_n^{(j)} e^{-n\lambda u},$$

setting $s = 1$ yields [equation \(3.3\)](#). The coefficients agree with the analytic solution by uniqueness. Cauchy's estimate on the z -disk and the bound $|U| \leq \kappa r$ give [equation \(3.4\)](#). $\square$

For $d = G'(v_0)$ and $g_2 = G''(v_0)$, the first two terms are

$$U_1 = -\frac{a_1}{d}, \quad (3.5)$$

$$U_2 = -\frac{a_2}{d} - \frac{\lambda a_1^2}{d^2} - \frac{g_2 a_1^2}{2d^3}. \quad (3.6)$$

Thus a leading exponential coefficient that cancels causes no division by zero: the formula never divides by a_1 .

For the elementary core

$$G(v) = av + b \operatorname{Log} v + c/v + d_0, \quad (3.7)$$

all U_n are rational functions of $v_0, a, b, c, \lambda, a_1, \dots, a_n$. The only possible finite denominator loci are $v_0 = 0$ and $G'(v_0) = 0$; the power of the latter denominator is at most $2n - 1$. Indeed, at most $j - 1$ derivatives fall on D^{-j} , which produces at most $D^{-(2j-1)}$. If L starts in degree ν , only $j \leq \lfloor n/\nu \rfloor$ contributes. This is a useful algebraic stability statement under cancellation.

If $a \neq 0$ in [equation \(3.7\)](#), then on a fixed u -disk, $D_{v_0}(u) \rightarrow a$ as $v_0 \rightarrow \infty$. The hypotheses therefore hold uniformly for large v_0 in a strict decay sector $\Re(\lambda v_0) \rightarrow +\infty$. When $a = 0$, that uniform assertion need not hold with the same disk and scaling. The next theorem addresses a different degeneration.

3.2 A regular finite endpoint with an essential correction

Let G be holomorphic near 0, $g_1 = G'(0) \neq 0$, and consider, on a strict decay sector,

$$F(t) = G(t) + L(e^{-S/t}), \quad L(0) = 0, \quad S \neq 0. \quad (3.8)$$

The exact inverse is close to the ordinary analytic inverse of G , but differentiating the exponential introduces t^{-2} . This determines the correct uniform scale.

Theorem 3.2 (Sharp finite-endpoint feedback scale). *Let $t_0 = G^{-1}(w)$ be the local core inverse and $Q_0 = e^{-S/t_0}$. There is a holomorphic function $U(t_0, \xi)$ on a fixed bidisk, with $U(t_0, 0) = 0$, such that for sufficiently small t_0 in any strict decay subsector the unique nearby inverse is*

$$t = t_0 + t_0^2 U\left(t_0, \frac{Q_0}{t_0^2}\right) = t_0 + \sum_{n \geq 1} d_n(t_0) Q_0^n. \quad (3.9)$$

Writing $U(t_0, \xi) = \sum_{n \geq 1} U_n(t_0) \xi^n$,

$$d_n(t_0) = t_0^{2-2n} U_n(t_0).$$

Hence d_n has pole order at most $2n - 2$. If $a_1 = L'(0) \neq 0$, this order is exact, and

$$U_n(0) = \frac{(-1)^n n^{n-1}}{n!} S^{n-1} \left(\frac{a_1}{g_1}\right)^n. \quad (3.10)$$

If $|U| \leq M$ on $|t_0| \leq r_0$, $|\xi| \leq R$, then for $\eta = |Q_0|/(R|t_0|^2) < 1$,

$$\left| t - t_0 - \sum_{n=1}^N d_n(t_0) Q_0^n \right| \leq |t_0|^2 \frac{M \eta^{N+1}}{1 - \eta}. \quad (3.11)$$

Proof. Set $t = t_0 + t_0^2 u$, and temporarily regard $\xi = Q_0/t_0^2$ as independent. The identity

$$e^{-S/t} = Q_0 \exp\left(\frac{Su}{1 + t_0 u}\right)$$

transforms $F(t) = G(t)$, after division by t_0^2 , into

$$\begin{aligned} 0 &= u \int_0^1 G'(t_0 + st_0^2 u) ds \\ &\quad + \xi \exp\left(\frac{Su}{1 + t_0 u}\right) L_1\left(t_0^2 \xi \exp\left(\frac{Su}{1 + t_0 u}\right)\right), \quad L_1(z) = L(z)/z. \end{aligned}$$

This is jointly holomorphic across $t_0 = 0$. Its derivative in u at the origin is g_1 . The implicit-function theorem gives U on a fixed bidisk. On a strict decay subsector, $Q_0/t_0^2 \rightarrow 0$, so the physical substitution is valid. The translated point remains in the sector after a harmless shrinking.

At $t_0 = 0$, the equation becomes

$$g_1 U(0, \xi) + a_1 \xi e^{SU(0, \xi)} = 0.$$

On the branch $W(0) = 0$,

$$U(0, \xi) = -\frac{1}{S} W\left(\frac{S a_1}{g_1} \xi\right). \quad (3.12)$$

Lagrange inversion for $W e^W = z$ gives $W(z) = \sum_{n \geq 1} (-n)^{n-1} z^n / n!$, proving equation (3.10). The displayed numbers are nonzero when $S a_1 g_1 \neq 0$. Finally, Cauchy's estimate in ξ , followed by multiplication by $|t_0|^2$, proves equation (3.11). $\square$

The first displacement is

$$t - t_0 = -\frac{a_1}{G'(t_0)}Q_0 + \mathcal{O}\left(\frac{|Q_0|^2}{|t_0|^2}\right).$$

For fixed N , the uniform next scale is $\mathcal{O}(|Q_0|^{N+1}/|t_0|^{2N})$, not a uniform $\mathcal{O}(|Q_0|^{N+1})$. The nonzero leading poles in [equation \(3.10\)](#) show why that distinction cannot be removed in general. The theorem extends to $L(t, z)$ holomorphic in two variables with $L(t, 0) = 0$; the limiting coefficient is then $\partial_z L(0, 0)$.

3.3 A complex residual certificate

A lower bound on $|F'|$ by itself is insufficient for a global complex inverse error estimate. For example, e^v has no zero derivative but takes equal values at points differing by $2\pi i$. The following stronger condition is elementary and usable.

Proposition 3.3 (Residual-to-root certificate). *Let $D = \{v : |v - \hat{v}| \leq r\}$, and suppose F is holomorphic on a neighborhood of D . If $a \neq 0$, $0 \leq \theta < 1$, and*

$$\sup_D |F'(v) - a| \leq \theta|a|, \quad |F(\hat{v}) - w| \leq (1 - \theta)|a|r,$$

then $F(v) = w$ has a unique root v_ in D , and*

$$|\hat{v} - v_*| \leq \frac{|F(\hat{v}) - w|}{(1 - \theta)|a|}.$$

For any $v_1, v_2 \in D$,

$$|F(v_1) - F(v_2)| \geq (1 - \theta)|a||v_1 - v_2|.$$

Proof. The map $T(v) = v - (F(v) - w)/a$ has Lipschitz constant at most θ on the convex disk. The residual assumption makes it a self-map of the closed disk. The contraction theorem gives its unique fixed point. Integration along the segment from v_1 to v_2 gives

$$F(v_1) - F(v_2) = a(v_1 - v_2) + \int_{v_2}^{v_1} (F'(v) - a) dv,$$

and the reverse triangle inequality proves the claimed lower bound and hence the residual estimate. $\square$

For $t = 1/v$, transporting the result uses the exact identity

$$\left| \frac{1}{v} - \frac{1}{v_*} \right| = \frac{|v - v_*|}{|v||v_*|}.$$

A certificate should therefore include the branch disk as well as the residual. This avoids applying a real mean-value argument where cancellation along a complex path is possible.

4 Cyclotomic Borel transforms and the necessary Stokes correction

The repository's conjecture `conj:cyclotomic-resurgent-inverse` [17] contains two different issues: resurgence of the normalized logarithm of the product, and identification of all its

directional sums with one fixed analytic function. The first issue admits an explicit answer throughout the stated disk $|a| < 1$. The second identification is false as written: different nonsingular Laplace rays can give different functions even where both integrals converge and the original product is defined. The correction is an explicitly computable exponentially small term.

This analysis belongs to an established literature. Marmi and Sauzin [5, §4.2–4.3, especially Theorem 4.3] study Borel transforms at arbitrary root-of-unity resonances for the cohomological equation. Their general solution $f_g(q, z) = \sum_{r \geq 1} g_r z^r / (q^r - 1)$ includes the present logarithm when $g(z) = -\log(1 - z)$. Their variable is $\eta = \log(q/\zeta) = -t$, and their theorem is formulated for ηf_g ; differentiation in the Borel variable gives the normalization used below. Garoufalidis and Kashaev [6] give explicit poles, residues, and Stokes transformations for Faddeev's quantum dilogarithm. The more recent work of Fantini and Rella [7, §3] studies q -Pochhammer resurgence and the correction terms needed to reconstruct particular q -series. Our purpose is a directly checkable specialization, with uniform error bounds and a concrete repair of the repository's normalization claim.

4.1 Normalization and the exact Borel transform

Let $m \geq 1$, let h be an integer coprime to m , and put $\zeta = \exp(2\pi i h/m)$. For $|a| < 1$ and $\Re t > 0$, define

$$L_a(t) = \log(a; \zeta e^{-t})_\infty = - \sum_{r=1}^{\infty} \frac{a^r}{r(1 - \zeta^r e^{-rt})}. \quad (4.1)$$

Here and below the logarithm is defined by its convergent series. Set

$$S_{-1}(a) = - \frac{\text{Li}_2(a^m)}{m^2}, \quad (4.2)$$

$$S_0(a) = \frac{1}{2m} \log(1 - a^m) - \sum_{\substack{r \geq 1 \\ m \nmid r}} \frac{a^r}{r(1 - \zeta^r)}, \quad (4.3)$$

$$R_a(t) = L_a(t) - \frac{S_{-1}(a)}{t} - S_0(a). \quad (4.4)$$

For $j \in \mathbb{Z} \setminus \{0\}$, let

$$\omega_j = \frac{2\pi i j}{m}, \quad p_j \in \{1, \dots, m\}, \quad h p_j \equiv j \pmod{m}. \quad (4.5)$$

The convention $p_j = m$ represents the zero residue class.

Theorem 4.1 (Explicit cyclotomic Borel transform). *The normalized function $R_a(t)$ has a formal expansion $\tilde{R}_a(t) = \sum_{n \geq 1} c_n(a) t^n$ with*

$$c_n(a) = \sum_{r \geq 1} a^r r^{n-1} \sum_{\substack{j \neq 0 \\ j \equiv h r \pmod{m}}} \omega_j^{-n-1} \quad (n \geq 1). \quad (4.6)$$

Its Borel transform, with the convention $\mathcal{B}(t^n) = \xi^{n-1}/(n-1)!$, is

$$B_{a,\zeta}(\xi) = \sum_{j \neq 0} \frac{1}{\omega_j^2} \frac{a^{p_j} \exp(p_j \xi / \omega_j)}{1 - a^m \exp(m \xi / \omega_j)}. \quad (4.7)$$

For $a \neq 0$, choose any λ with $a = e^{-\lambda}$; necessarily $\Re \lambda > 0$. The complete set of singularities is

$$\Sigma_{a,\zeta} = \left\{ \xi_{j,n} := \omega_j \left(\lambda + \frac{2\pi i n}{m} \right) : j \in \mathbb{Z} \setminus \{0\}, n \in \mathbb{Z} \right\}. \quad (4.8)$$

These points are distinct, nonzero, and locally finite. Every singularity is a simple pole, with

$$\operatorname{Res}_{\xi=\xi_{j,n}} B_{a,\zeta}(\xi) = -\frac{\exp(2\pi i n p_j/m)}{m\omega_j}. \quad (4.9)$$

Thus the Borel germ extends as a single-valued meromorphic function on $\mathbb{C}$, in particular along every path avoiding $\Sigma_{a,\zeta}$. For $a = 0$ it is identically zero.

Proof. For a fixed $r \geq 1$, let $J_r = \{j \neq 0 : j \equiv hr \pmod{m}\}$ and put

$$\kappa_r = \begin{cases} 1/2, & m \mid r, \\ (1 - \zeta^r)^{-1}, & m \nmid r. \end{cases}$$

The Mittag–Leffler expansion of the hyperbolic cotangent, after subtraction of its value at zero and removal of a possible pole at zero, gives

$$\frac{1}{1 - \zeta^r e^{-z}} = \frac{\mathbf{1}_{m \mid r}}{z} + \kappa_r - z \sum_{j \in J_r} \frac{1}{\omega_j^2 (1 - z/\omega_j)}. \quad (4.10)$$

For clarity, the nonconstant part of the partial-fraction expansion is $\sum_{j \in J_r} [(z - \omega_j)^{-1} + \omega_j^{-1}]$, which equals the last term in (4.10). This subtracted series converges normally on compact sets away from its poles. The identity also follows immediately from $1/(1 - e^{-w}) = 1/2 + (1/2) \coth(w/2)$ after a purely imaginary translation.

Substitute $z = rt$ in (4.10), multiply by $-a^r/r$, and sum over r . The pole and constant contributions are precisely (4.2) and (4.3). Consequently

$$R_a(t) = t \sum_{r \geq 1} a^r \sum_{j \in J_r} \frac{1}{\omega_j^2 (1 - rt/\omega_j)}. \quad (4.11)$$

All relevant series converge absolutely for each $\Re t > 0$; this follows, for example, from the denominator estimate in the next theorem with a closed subsector containing t . Expanding the rational factor formally gives (4.6). Applying the Borel transform gives

$$B_{a,\zeta}(\xi) = \sum_{r \geq 1} a^r \sum_{j \in J_r} \omega_j^{-2} e^{r\xi/\omega_j}.$$

In a sufficiently small disk about zero, these series converge absolutely. For a given j , the permitted r are $p_j + \ell m$, $\ell \geq 0$; their geometric sum is (4.7).

The denominator of its j th summand vanishes exactly when $\xi/\omega_j = \lambda + 2\pi i n/m$. At such a point the numerator is $\exp(2\pi i n p_j/m)$ and the derivative of the denominator is $-m/\omega_j$, proving (4.9). Moreover

$$\Im \xi_{j,n} = \frac{2\pi j}{m} \Re \lambda.$$

In a bounded disk this bounds $|j|$, and then bounds $|n|$. It also shows that equality of two locations forces equality first of j , then of n . In particular, there is no cancellation between summands at a pole. Finally the tail of (4.7) is locally $O(j^{-2})$, since $\xi/\omega_j \rightarrow 0$ and $1 - a^m \neq 0$. This proves normal convergence away from the displayed poles and completes the continuation argument. Changing the logarithm λ merely reindexes n . $\square$

Corollary 4.2 (Exact convergence radius and large-order coefficients). *For $a \neq 0$, the convergence radius of the Borel germ is exactly*

$$d(a, m) = \frac{2\pi}{m} \min_{n \in \mathbb{Z}} \left| \lambda + \frac{2\pi i n}{m} \right|. \quad (4.12)$$

It depends on the order m but not on the choice of primitive m th root ζ . The formal expansion has optimal Gevrey order one: it does not have Gevrey order $s < 1$. Let P_0 be the finite set of poles of modulus $d(a, m)$, let $d_1 > d(a, m)$ be the next pole modulus, and fix $d(a, m) < b < d_1$. Then

$$\frac{c_n(a)}{(n-1)!} = - \sum_{\sigma \in P_0} \frac{\text{Res}_{\xi=\sigma} B_{a,\zeta}(\xi)}{\sigma^n} + O(b^{-(n-1)}) \quad (n \rightarrow \infty). \quad (4.13)$$

The finite sum is kept intact; it may cancel on subsequences, as it must for the vanishing even coefficients when $m = 1$.

Proof. Minimizing (4.8) first over j gives $|j| = 1$, proving (4.12). None of these poles is removable by (4.9). If $|c_n| \leq CA^n(n!)^s$ for some $s < 1$, the Borel transform would have infinite convergence radius, a contradiction. Finally subtract from $B_{a,\zeta}$ the finitely many principal parts $\text{Res}_\sigma B/(\xi - \sigma)$ for $\sigma \in P_0$. The difference is holomorphic on $|\xi| < d_1$. Cauchy's coefficient bound on $|\xi| = b$ and the geometric expansion of each subtracted principal part give (4.13). $\square$

4.2 Uniform Gevrey bounds and reconstruction of the product

Theorem 4.3 (Uniform remainders and positive-ray reconstruction). *Fix $0 < \rho < 1$ and $0 < \delta < \pi/2$, and write $\ell = -\log \rho$. Uniformly for $|a| \leq \rho$ and $|\arg t| \leq \pi/2 - \delta$, for every integer $N \geq 1$,*

$$\left| R_a(t) - \sum_{n=1}^{N-1} c_n(a)t^n \right| \leq \frac{m\pi}{6\rho \sin \delta} \left(\frac{m}{2\pi\ell} \right)^N (N-1)! |t|^N. \quad (4.14)$$

The Borel transform is uniformly bounded on the real axis:

$$\sup_{\xi \in \mathbb{R}} |B_{a,\zeta}(\xi)| \leq \frac{m^2 \rho}{12(1 - \rho^m)}. \quad (4.15)$$

More generally it is holomorphic and uniformly bounded on every closed horizontal strip $|\Im \xi| \leq b$ with $b < 2\pi\ell/m$. For all $\Re t > 0$,

$$R_a(t) = \int_0^\infty e^{-\xi/t} B_{a,\zeta}(\xi) d\xi. \quad (4.16)$$

Proof. If $|\arg t| \leq \pi/2 - \delta$, then the argument of rt/ω_j is separated from the positive real axis by at least δ . Hence

$$|1 - rt/\omega_j| \geq \sin \delta \quad (r \geq 1, j \neq 0).$$

Use the finite geometric identity $(1 - u)^{-1} = \sum_{k=0}^{N-2} u^k + u^{N-1}(1 - u)^{-1}$ in (4.11); for $N = 1$ the finite sum is empty. The exact remainder equals

$$t^N \sum_{r \geq 1} a^r r^{N-1} \sum_{j \in J_r} \frac{\omega_j^{-N-1}}{1 - rt/\omega_j}.$$

Its absolute value is at most

$$\frac{|t|^N}{\sin \delta} \sum_{r \geq 1} \rho^r r^{N-1} \cdot 2\zeta(N+1) \left(\frac{m}{2\pi} \right)^{N+1}.$$

Here $\zeta(s)$ denotes the Riemann zeta function, not the root of unity. Since

$$\sum_{r \geq 1} e^{-\ell r} r^{N-1} \leq e^\ell \int_0^\infty e^{-\ell x} x^{N-1} dx = \frac{(N-1)!}{\rho \ell^N},$$

and $\zeta(N+1) \leq \zeta(2) = \pi^2/6$, the stated constant follows. The integral estimate can be checked termwise by integrating on $[r, r+1]$; enlarging $[1, \infty)$ to $[0, \infty)$ only increases it.

For real ξ , all exponentials e^{ξ/ω_j} have modulus one. Thus the absolute value of the rational factor in (4.7) is at most $\rho/(1-\rho^m)$, while $\sum_{j \neq 0} |\omega_j|^{-2} = m^2/12$. For the horizontal strip, use

$$|a^m e^{m\xi/\omega_j}| \leq \rho^m \exp\left(\frac{m^2 b}{2\pi|j|}\right) \leq \rho^m \exp\left(\frac{m^2 b}{2\pi}\right) < 1.$$

The numerator is uniformly bounded as well, proving the strip assertion.

Finally, on the positive real Borel ray the double series before the geometric summation is absolutely dominated by $\sum_r \rho^r \sum_{j \neq 0} |\omega_j|^{-2}$. Fubini's theorem and $\Re(1/t) > 0$ give

$$\int_0^\infty e^{-\xi/t} e^{r\xi/\omega_j} d\xi = \frac{t}{1 - rt/\omega_j}.$$

Equation (4.11) now proves (4.16). $\square$

4.3 Admissible rays and accumulation of Stokes directions

For a ray avoiding (4.8), define

$$\mathcal{S}_\theta \tilde{R}_a(t) = \int_0^{e^{i\theta}\infty} e^{-\xi/t} B_{a,\zeta}(\xi) d\xi. \quad (4.17)$$

For each fixed a and each such direction, the Borel transform is bounded on the ray, so this integral converges whenever $\Re(e^{i\theta}/t) > 0$. Indeed there are only finitely many profiles

$$F_p(w) = \frac{a^p e^{pw}}{1 - a^m e^{mw}}, \quad p = 1, \dots, m,$$

and two ray orientations for $w = \xi/\omega_j$. If $\Re w \rightarrow -\infty$ the profile tends to zero. If $\Re w \rightarrow +\infty$, it is asymptotic to $-a^{p-m} e^{(p-m)w}$, which is bounded because $p \leq m$. If $\Re w = 0$, the denominator is bounded below by $1 - |a|^m$. In the remaining compact part of each profile ray, avoidance of its zeros gives a positive minimum for the denominator. Summing $O(|\omega_j|^{-2})$ proves the assertion. No Diophantine condition is being tacitly assumed; the bound is for fixed a and a fixed nonsingular ray, and need not stay bounded when the ray or parameter approaches a singular configuration.

This raywise statement requires a distinction from a theorem asserting analyticity in an infinite angular Borel sector. For fixed $a \neq 0$, the directions of (4.8) accumulate at both real directions as $n \rightarrow \pm\infty$. In particular, although the positive real ray itself is nonsingular and (4.16) holds, there is no open angular sector about that ray which is free of all Borel poles. Its horizontal-strip continuation is the relevant substitute. The poles are locally finite in the Borel plane; it is their *directions* that accumulate, at points tending to infinity. These statements are compatible, and they should not be conflated with the assumptions on a finite collection of actions used in a particular transseries chart.

Whenever a deformation from an admissible contour γ_- to γ_+ sweeps the poles in a set P , with vanishing closing integrals and an absolutely convergent residue sum, the residue theorem gives the exact identity

$$\mathcal{S}_+ \tilde{R}_a(t) - \mathcal{S}_- \tilde{R}_a(t) = \sum_{(j,n): \xi_{j,n} \in P} \frac{e^{2\pi i n p_j/m}}{j} e^{-\xi_{j,n}/t}. \quad (4.18)$$

The orientation convention is that the contour formed by γ_- followed by $-\gamma_+$ encloses P counterclockwise. Equivalently, its right side is $-2\pi i \sum_{\xi \in P} \text{Res}_\xi [e^{-\xi/t} B_{a,\zeta}(\xi)]$. The contour hypotheses are stated explicitly; the next proposition checks them with elementary estimates in the case needed to disprove the repository's assertion.

4.4 A concrete counterexample to direction-independent normalization

Proposition 4.4 (Two nonsingular rays give different sums). *Take $m = 1$, $\zeta = 1$, and $a = e^{-1}$. Both the positive real ray and the ray of angle $\pi/4$ avoid every Borel singularity, and their Laplace integrals converge for every $t > 0$. They satisfy*

$$\mathcal{S}_{\pi/4} \tilde{R}_a(t) - R_a(t) = \sum_{j=1}^{\infty} \sum_{n=1}^{\infty} \frac{1}{j} \exp \left[-\frac{j(4\pi^2 n + 2\pi i)}{t} \right] \quad (4.19)$$

$$= - \sum_{n=1}^{\infty} \log \left(1 - \exp \left[-\frac{4\pi^2 n + 2\pi i}{t} \right] \right). \quad (4.20)$$

The logarithms in the second line are their convergent power-series branches. The difference is nonzero for every $t = 1/N$, $N \geq 1$ an integer, and for all sufficiently small positive t .

Proof. For $m = 1$, pairing the summands j and $-j$ in (4.7) gives

$$B_{a,1}(\xi) = -\frac{1}{4\pi^2} \sum_{j=1}^{\infty} \frac{1}{j^2} \left(\frac{ae^{i\xi/(2\pi j)}}{1 - ae^{i\xi/(2\pi j)}} + \frac{ae^{-i\xi/(2\pi j)}}{1 - ae^{-i\xi/(2\pi j)}} \right). \quad (4.21)$$

The poles in the open wedge $0 < \arg \xi < \pi/4$ are precisely

$$\xi = 4\pi^2 jn + 2\pi i j, \quad j, n \geq 1,$$

each with residue $i/(2\pi j)$. No pole lies on either boundary ray: in the upper half-plane a pole on the diagonal would require $4\pi^2 jn = 2\pi j$, or $2\pi n = 1$.

To justify contour rotation despite the infinitely many Stokes directions near zero, rotate *each paired j th summand* first. Put $\xi = 2\pi j u$. Close the sector from the positive real u -axis to the diagonal along the vertical line $\Re u = (2M + 1)\pi$, with $M \rightarrow \infty$. On this vertical segment both numbers ae^{iu} and ae^{-iu} are negative real. Each quotient $ae^{\pm iu}/(1 - ae^{\pm iu})$ therefore has modulus at most one. For $t > 0$, the modulus of the exponential kernel there is $\exp[-2\pi j(2M + 1)\pi/t]$; the segment length grows only linearly in M . The closing integral consequently tends to zero.

The residue theorem, with the indicated orientation, gives for this paired summand

$$(\mathcal{S}_{\pi/4} - \mathcal{S}_0)_j = \frac{1}{j} \sum_{n \geq 1} e^{-j(4\pi^2 n + 2\pi i)/t}.$$

The integrals over either infinite ray can be summed over j by the bounded-profile estimate and $\sum j^{-2} < \infty$; the displayed residue sums are absolutely convergent. Summing proves (4.19), because $\mathcal{S}_0 \tilde{R}_a = R_a$ by (4.16). Summation first over j gives (4.20).

For $t = 1/N$ all factors $e^{-2\pi i j/t}$ equal one, so every term in (4.19) is positive real. More generally, as $t \downarrow 0$,

$$\mathcal{S}_{\pi/4} \tilde{R}_a(t) - R_a(t) = e^{-(4\pi^2 + 2\pi i)/t} (1 + O(e^{-4\pi^2/t})),$$

since every other pair (j, n) has $jn \geq 2$. This also proves nonvanishing for all sufficiently small positive t . $\square$

Consequently the conjecture's uniform Gevrey assertion, meromorphic continuation assertion, and positive-ray reconstruction are valid in the stated fixed-parameter disk. Its demand that *every* nonsingular convergent directional sum equal the single fixed product is refuted. One can either keep the positive-ray normalization or attach the explicit Stokes correction to each other contour. Keeping only the algebraic expansion cannot distinguish these functions.

4.5 The corresponding correction for an inverse branch

The preceding computation does not, by itself, establish every functional analytic hypothesis in a parameter-dependent resurgent implicit-function theorem. It does establish the exact transformation that any such theorem must respect. Suppose on a common sector that

$$F_+(a, t) = F_-(a, t) + D(a, t),$$

where $F_\pm$ are two sectorial determinations of L_a and D is the residue correction. Let $v_\pm(t)$ be the corresponding sectorial data, and let $a_\pm(t)$ be chosen local solutions of

$$F_\pm(a_\pm, t) = F_\pm(a_0, t) + v_\pm(t).$$

Set

$$\begin{aligned} H_-(a, t) &= F_-(a, t) - F_-(a_0, t) - v_-(t), \\ E(a, t) &= D(a, t) - D(a_0, t) - (v_+(t) - v_-(t)). \end{aligned}$$

Then the exact relation is

$$H_-(a_+(t), t) = -E(a_+(t), t). \quad (4.22)$$

If $A_-(w, t)$ denotes the chosen local inverse of $a \mapsto H_-(a, t)$, this is equivalently

$$a_+(t) = A_-(-E(a_+(t), t), t), \quad a_-(t) = A_-(0, t). \quad (4.23)$$

Existence and uniqueness here are local analytic statements, requiring nonvanishing of $\partial_a H_-$ and sufficiently small perturbations within the chosen inverse chart.

For a precise first-order version, insert a scalar parameter ϵ and solve $H_-(a_\epsilon, t) + \epsilon E(a_\epsilon, t) = 0$ near $\epsilon = 0$. Differentiating the ordinary implicit equation gives

$$\left. \frac{\partial a_\epsilon(t)}{\partial \epsilon} \right|_{\epsilon=0} = -\frac{E(a_-(t), t)}{\partial_a H_-(a_-(t), t)}. \quad (4.24)$$

This formulation avoids an unproved uniform error estimate when either derivatives or inverse radii depend on t . It also makes the normalization essential: the correction at the reference point a_0 and any jump in the target v must both be included.

For inversion with a fixed target y rather than a subtracted reference value, the same statement reads

$$F_+(\cdot, t)^{-1} = (F_-(\cdot, t) + D(\cdot, t))^{-1}$$

on the chosen local branch. Thus, whenever a resurgent algebra and its summation maps satisfy the required composition and implicit-function closure hypotheses, the Stokes map acts on the *equation and its target*, and the transformed equation is then inverted. Inversion does not remove the Stokes correction.

5 Exact modular normal forms at rational cusps

The modular formula below is classical. Its role here is to provide an exact analytic representative, including phases, whose inverse can then be studied without losing a flat term. The eta transformation and the Dedekind-sum multiplier are given in [9, Equations 23.18.5–23.18.7]; an elementary proof of the transformation is available in [8].

5.1 One factor with all phases specified

For integers H, K , $K > 0$, $(H, K) = 1$, write

$$s(H, K) = \sum_{\ell=1}^{K-1} \frac{\ell}{K} \left(\frac{H\ell}{K} - \left\lfloor \frac{H\ell}{K} \right\rfloor - \frac{1}{2} \right), \quad s(H, 1) = 0.$$

Fix a cusp h/k , and for each positive integer d put

$$g_d = (d, k), \quad H_d = \frac{dh}{g_d}, \quad K_d = \frac{k}{g_d}.$$

Choose a_d with $a_d H_d \equiv -1 \pmod{K_d}$, and define

$$\rho_d = e^{2\pi i a_d / K_d}.$$

For $K_d = 1$, set $\rho_d = 1$. The value of ρ_d is independent of the chosen representative of a_d .

Lemma 5.1 (One-factor cusp identity). *If $\tau = h/k + it/(2\pi)$ and $\Re t > 0$, then*

$$\begin{aligned} \eta(d\tau) = & \exp \left(\pi i \left[\frac{dh}{12k} - s(H_d, K_d) \right] \right) \left(\frac{2\pi g_d}{kdt} \right)^{1/2} \\ & \times \exp \left(-\frac{\pi^2 g_d^2}{6dk^2 t} \right) P \left(\rho_d \exp \left(-\frac{4\pi^2 g_d^2}{dk^2 t} \right) \right). \end{aligned} \quad (5.1)$$

The square root is the branch positive for $t > 0$.

Proof. Choose $b_d \in \mathbb{Z}$ so that

$$\gamma_d = \begin{pmatrix} a_d & b_d \\ K_d & -H_d \end{pmatrix} \quad \text{has determinant } 1.$$

A direct calculation gives

$$\gamma_d(d\tau) = \frac{a_d}{K_d} + \frac{2\pi i}{K_d^2 dt}.$$

The eta multiplier, in the convention of [9], is

$$\epsilon(\gamma_d) = \exp \left(\pi i \left[\frac{a_d - H_d}{12K_d} + s(H_d, K_d) \right] \right),$$

and

$$\eta(\gamma_d z) = \epsilon(\gamma_d) [-i(K_d z - H_d)]^{1/2} \eta(z).$$

At $z = d\tau$, the square root equals $(K_d dt/(2\pi))^{1/2}$. Insert the product definition of eta at $\gamma_d(d\tau)$ and solve for $\eta(d\tau)$. The a_d phase cancels, leaving $\pi i H_d/(12K_d) - \pi i s(H_d, K_d)$. Since $H_d/K_d = dh/k$, this is [equation \(5.1\)](#). All transformations take place in the upper half-plane: $\Re t > 0$ implies $\Re(1/t) > 0$, so the dual product converges. $\square$

5.2 A finite product quotient

Let $\mathcal{D}$ be a finite set of distinct positive integers with nonzero integer exponents r_d . Set

$$\begin{aligned} B &= \sum_{d \in \mathcal{D}} r_d, & D &= \sum_{d \in \mathcal{D}} dr_d, \\ A_{h,k} &= \frac{\pi^2}{6k^2} \sum_{d \in \mathcal{D}} \frac{r_d g_d^2}{d}, & N &= \text{lcm}\{d : d \in \mathcal{D}\}, \\ S &= \frac{4\pi^2}{k^2 N}, & M_d &= \frac{N g_d^2}{d}, \quad Q = e^{-S/t}. \end{aligned} \quad (5.2)$$

Each M_d is a positive integer. Define

$$R(Q) = \prod_{d \in \mathcal{D}} P(\rho_d Q^{M_d})^{r_d}, \quad (5.3)$$

and the nonzero constant

$$\kappa_{h,k} = \prod_{d \in \mathcal{D}} \left(\frac{2\pi g_d}{kd} \right)^{r_d/2} \exp \left(-\pi i \sum_{d \in \mathcal{D}} r_d s(H_d, K_d) \right). \quad (5.4)$$

Theorem 5.2 (Exact finite-product cusp normal form). *For $q = e^{2\pi i h/k-t}$, $\Re t > 0$,*

$$\prod_{d \in \mathcal{D}} P(q^d)^{r_d} = \kappa_{h,k} t^{-B/2} \exp \left(-\frac{A_{h,k}}{t} + \frac{Dt}{24} \right) R(Q). \quad (5.5)$$

For the eta quotient,

$$\prod_{d \in \mathcal{D}} \eta(d\tau)^{r_d} = \kappa_{h,k} e^{\pi i h D/(12k)} t^{-B/2} \exp \left(-\frac{A_{h,k}}{t} \right) R(Q). \quad (5.6)$$

These are exact holomorphic identities, not finite asymptotic expansions.

Proof. Multiply [equation \(5.1\)](#) with the indicated integer powers. The equality $g_d^2/d = M_d/N$ puts all exponentials on the common grid Q^{M_d} . For the raw product, use

$$P(q^d) = e^{-\pi i d \tau/12} \eta(d\tau).$$

This removes the phase $e^{\pi i d h/(12k)}$ and introduces $e^{dt/24}$. The finite products of the remaining constants give [equation \(5.4\)](#). Negative integer powers cause no difficulty because each factor is nonzero in its disk. $\square$

At $q = 1$, the formula reduces to

$$P(e^{-dt}) = \sqrt{\frac{2\pi}{dt}} e^{-\pi^2/(6dt) + dt/24} P(e^{-4\pi^2/(dt)}).$$

The extension to a rational cusp is not just replacement of t by $k^2 t$: the factors g_d and phases ρ_d also change.

5.3 Arithmetic coefficients and resonant cancellation

The logarithm of the dual product is convergent for $|Q| < 1$:

$$\begin{aligned} L(Q) &= \log R(Q) = \sum_{n \geq 1} \lambda_n Q^n, \\ \lambda_n &= - \sum_{\substack{d \in \mathcal{D} \\ M_d | n}} r_d \sigma_{-1}(n/M_d) \rho_d^{n/M_d}. \end{aligned} \quad (5.7)$$

This follows by expanding $\log P(z) = -\sum_{n \geq 1} \sigma_{-1}(n) z^n$. If $R(Q) = \sum_{n \geq 0} c_n Q^n$, then

$$c_0 = 1, \quad n c_n = \sum_{j=1}^n j \lambda_j c_{n-j}. \quad (5.8)$$

Thus all coefficients belong to the cyclotomic number field generated by the finitely many ρ_d . No numerical cancellation test is needed.

The first nonzero coefficient of $R - 1$ equals the first nonzero coefficient of L . It must be found *after* the finite sum in [equation \(5.7\)](#), because distinct factors can cancel at the same action. For arbitrary phase-product representations the cancellation can even be complete. For example, separating even and odd factors gives

$$P(-Q) = \frac{P(Q^2)^3}{P(Q)P(Q^4)}.$$

The constant case must therefore be included whenever a general phase-product input is allowed.

For $0 < r < 1$,

$$\sup_{|Q| \leq r} |\log R(Q)| \leq \sum_{d \in \mathcal{D}} |r_d| \frac{r^{M_d}}{(1 - r^{M_d})^2}. \quad (5.9)$$

Indeed $\sum_{\ell, n \geq 1} |z|^{\ell n} / \ell \leq |z| / (1 - |z|)^2$. Together with Cauchy's formula, this supplies explicit coefficient and tail estimates.

5.4 The four cusp regimes

Taking logarithms of [equation \(5.5\)](#) and setting $v = 1/t$ gives

$$\log F = -A_{h,k}v + \frac{B}{2} \operatorname{Log} v + \frac{D}{24v} + \log \kappa_{h,k} + L(e^{-Sv}). \quad (5.10)$$

This is the exact-core form of [equation \(3.7\)](#). A local logarithm of the nonzero target is understood.

Condition	Perturbative behavior	Appropriate inverse coordinate
$A_{h,k} \neq 0$	Exponential–power core	$v = 1/t$, with a linear–logarithmic or Lambert core
$A_{h,k} = 0, B \neq 0$	Power of t	A chosen algebraic power branch, then exponential corrections
$A_{h,k} = B = 0, D \neq 0$, raw products	Nonconstant analytic background	The finite-endpoint theorem after taking logarithms
$A_{h,k} = B = D = 0$, raw products	Constant plus flat terms	Puiseux inverse in Q , then a logarithm

For eta quotients the $D/(24v)$ term is absent, so $A_{h,k} = B = 0$ already gives the last regime. This difference is why a normalization cannot be omitted when identifying a flat critical value.

For completeness, the equation $av + b \operatorname{Log} v = w - d_0$, when $ab \neq 0$, has the local expression

$$v = \frac{b}{a} W_j \left(\frac{a}{b} e^{(w-d_0)/b} \right),$$

with the W -branch chosen to satisfy the original logarithm equation. Exponentiating an equation can otherwise introduce a spurious sheet. A nonzero c/v can be retained in the exact core or treated by a separate analytic core correction.

The common nome in [equation \(5.2\)](#) is convenient and need not be primitive. Replacing Q by Q^g changes a displayed vanishing order. The constructions below have nonzero consecutive

exponents $m, m+1$, so their ramification cannot be removed by such a rescaling. All ramification statements elsewhere are relative to the stated coordinate Q .

Finally, for a fixed family of dilations the physical condition $Q \rightarrow 0$ is exactly

$$\frac{\Re(1/t)}{k^2} \longrightarrow +\infty.$$

On the real radial path this is $k^2 t \rightarrow 0$. A fixed-cusp expansion therefore does not imply a uniform small-nome approximation when the denominator k grows as fast as $t^{-1/2}$.

6 Convergent inversion at a flat critical value

The key analytic problem after complete cancellation is

$$F(t) = C R(e^{-S/t}), \quad R(Q) = 1 + c_\nu Q^\nu H(Q), \quad H(0) = 1, \quad (6.1)$$

where $C \neq 0$, $S > 0$, $c_\nu \neq 0$, and $\nu \geq 1$. The function R is holomorphic near zero. The output approaches C faster than every power of t on a strict right subsector. The finite multiplicity is a property of the *nome map* $Q \mapsto R(Q)$, not a finite Taylor multiplicity of F at the boundary point $t = 0$.

6.1 All coefficients and all local sheets

Put

$$Y = \frac{y/C - 1}{c_\nu}.$$

Choose the analytic root $H(Q)^{1/\nu}$ with value 1 at zero. Then $\chi(Q) = QH(Q)^{1/\nu}$ satisfies $\chi(0) = 0$, $\chi'(0) = 1$, and hence has an analytic inverse $Q = \Psi(u)$.

Theorem 6.1 (Complete small-nome flat inverse). *There is a disk on which Ψ is holomorphic, $\Psi(0) = 0$, $\Psi'(0) = 1$. Its coefficients and those of $\ell(u) = \log(\Psi(u)/u)$ are*

$$\Psi(u) = \sum_{n \geq 1} \frac{u^n}{n} [z^{n-1}] H(z)^{-n/\nu}, \quad (6.2)$$

$$\ell(u) = \sum_{n \geq 1} \frac{u^n}{n} [z^n] H(z)^{-n/\nu}. \quad (6.3)$$

For $Y \neq 0$ sufficiently small, fix a lift $\text{Log } Y$. Every inverse with sufficiently small $|e^{-S/t}|$ is exactly

$$u_j = \exp\left(\frac{\text{Log } Y + 2\pi i j}{\nu}\right), \quad t_j(Y) = -\frac{S}{(\text{Log } Y + 2\pi i j)/\nu + \ell(u_j)}, \quad j \in \mathbb{Z}. \quad (6.4)$$

These branches lie in $\Re t > 0$ when $|Y|$ is sufficiently small. A positive loop about $Y = 0$ sends j to $j + 1$, and

$$t_{j+\nu} = \frac{t_j}{1 - 2\pi i t_j / S}. \quad (6.5)$$

Proof. The analytic inverse-function theorem gives Ψ . Its equation is $\Psi = uH(\Psi)^{-1/\nu}$. Lagrange–Bürmann proves [equation \(6.2\)](#). Applying the same formula to $\ell(u) = -(1/\nu) \log H(\Psi(u))$, or differentiating $H(z)^{-n/\nu}$ inside the coefficient formula, gives [equation \(6.3\)](#).

The equation $R(Q) = y/C$ is equivalent to $\chi(Q)^\nu = Y$. It has exactly the ν roots $\Psi(u_j)$, with j modulo ν , in a sufficiently small Q -disk. Solving $e^{-S/t} = \Psi(u_j)$ requires all logarithm lifts. Since

$$\log \Psi(u_j) = \frac{\text{Log } Y + 2\pi i j}{\nu} + \ell(u_j)$$

is already a coherent lift, allowing j to range over all integers incorporates both the finite roots and all logarithmic lifts exactly once. Increasing j by ν leaves u_j unchanged and adds $2\pi i$ to the denominator. This proves [equation \(6.5\)](#) and the monodromy statement.

The real part of the denominator is $\log |\Psi(u_j)| < 0$ for small Y ; hence $-S$ divided by it has positive real part. Completeness is asserted in the small-nome region, where all roots of the analytic equation have just been counted. $\square$

The distinction in the last sentence matters. The theorem does not classify every possible tangential approach to $t = 0$ for which $|Q|$ remains near 1. Its natural cusp region is $\Re(S/t) > T$, with T sufficiently large.

6.2 The resulting transseries and its support

On one logarithm lift, put $L = -\log u$. Then

$$t = \frac{S}{L - \ell(u)} = \frac{S}{L} + S \sum_{n \geq 1} u^n \sum_{a=1}^n \frac{[u^n] \ell(u)^a}{L^{a+1}}, \quad (6.6)$$

for sufficiently small u . A sufficient condition for the regrouped series is $\sum_{n \geq 1} |[u^n] \ell(u)| |u|^n < |L|$. This is a convergent logarithmic–Puiseux representation. For each u^n , the coefficient is a finite polynomial in L^{-1} , with no hidden infinite inner sum. Since $u = Y^{1/\nu}$ on a chosen sheet, the power scale in the output variable is rationally ramified.

Let $\mathcal{A}$ be the additive monoid generated by the positive integers in the support of $H - 1$. [Equations \(6.2\)](#) and [\(6.3\)](#) show

$$\text{supp } \ell \subseteq \mathcal{A}.$$

Indeed every coefficient of $H^{-n/\nu}$ is a finite sum of products of coefficients of $H - 1$. Consequently no exponent outside $\mathcal{A}$ can occur in ℓ . This also shows which actions must be re-examined after cancellation: they are the *gaps above the first surviving action*, encoded by H , not a list of the ungrouped input factors.

6.3 An explicit uniform error bound

Theorem 6.2 (Certified logarithmic–Puiseux truncation). *Suppose H is holomorphic on a neighborhood of $|Q| \leq r$ and*

$$|H(Q) - 1| \leq \frac{1}{2} \quad (|Q| \leq r).$$

Put

$$R_0 = r 2^{-1/\nu}, \quad M = \frac{\log 2}{\nu}.$$

Then Ψ and ℓ are holomorphic for $|u| < R_0$, and $|\ell(u)| \leq M$. If ℓ_N is the Taylor polynomial through degree N , $\rho = |u|/R_0 < 1$, then

$$|\ell(u) - \ell_N(u)| \leq \frac{M\rho^{N+1}}{1 - \rho}. \quad (6.7)$$

For any coherent logarithm lift $L = -\log u$, assume $\rho \leq 1/2$ and $|L| \geq 2M$. The approximation $t_N = S/(L - \ell_N(u))$ satisfies

$$|t - t_N| \leq \frac{4SM\rho^{N+1}}{(1 - \rho)|L|^2}. \quad (6.8)$$

Proof. The disk $|H - 1| \leq 1/2$ excludes zero and defines the chosen root and logarithm. On $|Q| = r$,

$$|QH(Q)^{1/\nu}| \geq r 2^{-1/\nu} = R_0.$$

Rouché's theorem counts exactly one simple root of $QH(Q)^{1/\nu} = u$ in $|Q| < r$ for $|u| < R_0$. This extends Ψ analytically over the entire u -disk. Since

$$\ell(u) = -\frac{1}{\nu} \log H(\Psi(u)),$$

the power series for $\log(1 + z)$ gives $|\ell| \leq (\log 2)/\nu = M$. Cauchy's estimate proves [equation \(6.7\)](#).

There is no constant term in ℓ . Therefore, when $\rho \leq 1/2$,

$$|\ell_N(u)| \leq M \sum_{n=1}^N \rho^n \leq \frac{M\rho}{1 - \rho} \leq M.$$

Both $L - \ell$ and $L - \ell_N$ have modulus at least $|L|/2$. Subtracting their reciprocals proves [equation \(6.8\)](#). $\square$

For all the lifts in [equation \(6.4\)](#),

$$\Re(-\log u_j) = \frac{1}{\nu} \log(1/|Y|).$$

Thus the absolute error estimate can be made uniform over all integer sheet indices in the small-nome region. This does not say that all these points remain in one fixed angular subsector of t . As $|j|$ grows at fixed Y , the points approach 0 tangentially. If a fixed angular subsector is required, it imposes a bound of order $\log(1/|Y|)$ on the admissible magnitude of j .

6.4 Flatness, criticality, and nonconstancy

On a closed right subsector, $|Q| \leq e^{-S \sin \delta/|t|}$, so $F(t) - C = o(|t|^N)$ for every N . A nonconstant function of this form cannot extend holomorphically to a neighborhood of $t = 0$: such an extension would have every Taylor coefficient of $F - C$ equal to zero, hence would be constant. This explains why ordinary analytic ramification at $t = 0$ does not describe the problem.

A nonzero finite eta exponent vector with $A_{h,k} = B = 0$ cannot produce a constant function. If it did, analytic continuation in the upper half-plane would make the original eta quotient constant everywhere. As $q \rightarrow 0$ at infinity it is

$$q^{D/24} \prod_d P(q^d)^{r_d}.$$

For $D \neq 0$ this is not constant. For $D = 0$, let d_* be the smallest index with $r_{d_*} \neq 0$. The coefficient at q^{d_*} is $-r_{d_*} \neq 0$. This contradiction proves that a first nonzero dual coefficient exists in the balanced eta case. The analogous argument applies to a nonzero raw product satisfying all three balances.

The derivative of the inverse near the flat value is highly sensitive. From its leading term,

$$t \sim \frac{\nu S}{\log(1/Y)}, \quad \frac{dt}{dY} \sim \frac{\nu S}{Y \log^2(1/Y)}$$

on a fixed sheet. A tiny absolute error in the measured flat defect may therefore destroy useful relative information. Keeping the exact modular normalization and using logarithmic variables are part of the mathematics of stable inversion.

7 Sharp constructions with arbitrary ramification

The following results answer a concrete existence question: can complete cancellation of the perturbative cusp contribution coexist with arbitrarily high inverse ramification? The answer is yes, with a minimal number of factors. The minimality statements are relative to finite products on distinct positive dilations with nonzero integer exponents.

7.1 Three eta factors realize every order

Theorem 7.1 (Minimal three-factor eta construction). *For each integer $m \geq 2$, put*

$$\begin{aligned} K_m &= \frac{(m+1)^{(m^2-1)/2}}{m^{m^2/2}}, & S_m &= \frac{4\pi^2}{m(m+1)}, \\ F_m(t) &= K_m \frac{\eta(it/(2\pi))\eta(i(m+1)t/(2\pi))^{m^2-1}}{\eta(imt/(2\pi))^{m^2}}. \end{aligned} \quad (7.1)$$

For $\Re t > 0$,

$$F_m(t) = R_m(Q), \quad Q = e^{-S_m/t}, \quad R_m(Q) = \frac{P(Q^{m(m+1)})P(Q^m)^{m^2-1}}{P(Q^{m+1})^{m^2}}. \quad (7.2)$$

The expansion begins

$$R_m(Q) = 1 - (m^2 - 1)Q^m + m^2Q^{m+1} + \mathcal{O}(Q^{m+2}). \quad (7.3)$$

Thus the nome map has ramification order exactly m , and its support is primitive. The same order occurs at every rational cusp whose denominator is coprime to $m(m+1)$, after the explicit constant and phase normalization of [equation \(5.6\)](#). Three distinct eta factors are necessary for a nonzero exponent vector satisfying both $\sum r_d = 0$ and $\sum r_d/d = 0$.

Proof. On the dilations $1, m, m+1$, the exponent vector is

$$(r_1, r_m, r_{m+1}) = (1, -m^2, m^2 - 1).$$

Its two balances are

$$1 - m^2 + (m^2 - 1) = 0, \quad 1 - m + \frac{m^2 - 1}{m+1} = 0.$$

Applying [equation \(5.6\)](#) at $h = 0, k = 1$ cancels the power and essential exponential. The remaining constant is K_m^{-1} , proving [equation \(7.2\)](#). The factor with smallest dual exponent is $P(Q^m)^{m^2-1}$; the next is $P(Q^{m+1})^{-m^2}$. For $m \geq 2$, $2m \geq m + 2$, so no nonlinear product interferes with their first two coefficients. This proves [equation \(7.3\)](#). Since $m, m + 1$ both occur, their gcd is one.

If $(k, m(m + 1)) = 1$, all $g_d = 1$, and the same two balances hold. The dual exponents remain $m(m + 1), m + 1, m$. The leading coefficients become $-(m^2 - 1)\rho_{m+1}$ and $m^2\rho_m$. Both are nonzero, so the order and primitive support persist.

With only two distinct dilations d_1, d_2 , the first balance forces $r_2 = -r_1$, and the second gives $r_1(1/d_1 - 1/d_2) = 0$. Thus both exponents vanish. One factor is even more immediate. The three-factor construction attains the bound. $\square$

For completeness, the unramified order 1 is realized by

$$\sqrt{2} \frac{\eta(it/(2\pi))\eta(4it/(2\pi))^2}{\eta(2it/(2\pi))^3} = \frac{P(Q^4)P(Q)^2}{P(Q^2)^3} = 1 - 2Q + \mathcal{O}(Q^2), \quad Q = e^{-\pi^2/t}.$$

The exponent vector $(1, -3, 2)$ on dilations $(1, 2, 4)$ satisfies the same two balances.

There is no assertion here that every member of [equation \(7.1\)](#) has trivial modular character on $\Gamma_0(m(m + 1))$. In fact

$$\sum dr_d = m(m - 1)$$

need not be divisible by 24. The eta multiplier transformation is sufficient for the proof; the functions are analytic eta quotients with the stated multipliers.

For the normalized output

$$Y = \frac{1 - F_m(t)}{m^2 - 1}, \quad u^m = Y,$$

the first inverse coefficient is

$$Q = u + \frac{m}{m^2 - 1}u^2 + \mathcal{O}(u^3).$$

Every subsequent coefficient is obtained by [equation \(6.2\)](#). The principal radial branch has

$$t \sim \frac{mS_m}{\log(1/Y)} = \frac{4\pi^2/(m + 1)}{\log(1/Y)}.$$

Thus an unbounded finite ramification order coexists with a logarithmic rather than an algebraic leading inverse in t .

7.2 The smallest nontrivial eta example

For $m = 2$,

$$F_2(t) = \frac{3\sqrt{3}}{4} \frac{\eta(it/(2\pi))\eta(3it/(2\pi))^3}{\eta(2it/(2\pi))^4}, \quad Q = e^{-2\pi^2/(3t)}.$$

The exact dual series is

$$\begin{aligned} R_2(Q) = & 1 - 3Q^2 + 4Q^3 - 12Q^5 + 18Q^6 - 39Q^8 + 56Q^9 \\ & - 108Q^{11} + 148Q^{12} + \mathcal{O}(Q^{14}). \end{aligned} \tag{7.4}$$

With $s = ((1 - F_2)/3)^{1/2}$, the branch $Q \sim s$ is

$$Q = s + \frac{2}{3}s^2 + \frac{10}{9}s^3 + \frac{10}{27}s^4 - \frac{17}{27}s^5 - \frac{1168}{243}s^6 - \frac{4663}{486}s^7 - \frac{11086}{729}s^8 - \frac{48499}{13122}s^9 + \frac{879718}{19683}s^{10} + \mathcal{O}(s^{11}). \quad (7.5)$$

The corresponding logarithmic correction is

$$\ell(s) = \log(Q/s) = \frac{2}{3}s + \frac{8}{9}s^2 - \frac{22}{81}s^3 - \frac{85}{81}s^4 - \frac{1666}{405}s^5 - \frac{25199}{4374}s^6 + \mathcal{O}(s^7). \quad (7.6)$$

Consequently

$$t = \frac{2\pi^2/3}{-\log s - \ell(s)}.$$

Replacing s by $-s$ gives the second nome branch; the logarithm lifts supply the remaining sheets. All displayed rational coefficients were also checked by direct composition, independently of the analytic proof.

The type depends on the cusp. For the same normalized eta quotient, at $\tau = 1/2 + it/(2\pi)$, the exact identity is

$$\frac{3\sqrt{3}}{4} \frac{\eta(\tau)\eta(3\tau)^3}{\eta(2\tau)^4} = \frac{e^{\pi i/12}}{4} e^{\pi^2/(4t)} \frac{P(-Q^6)P(-Q^2)^3}{P(Q^{12})^4}, \quad Q = e^{-\pi^2/(6t)}. \quad (7.7)$$

The dual factor begins $1+3Q^2+\dots$. Hence the flat finite cusp at $q = 1$ becomes an exponentially growing cusp at $q = -1$. This is a direct illustration of why the arithmetic factors g_d in the normal form matter.

7.3 Four raw Euler-product factors: all perturbative terms cancel

For an unnormalized Euler product, two balances leave an $e^{Dt/24}$ factor. The following construction removes that term too, without inserting a compensating exponential by hand.

Theorem 7.2 (Minimal four-factor construction at every order). *Let $m \geq 1$,*

$$g = \gcd(m, 3), \quad c_j = m + j, \quad e_j = \frac{(-1)^j \binom{3}{j} (m + j)}{g} \quad (0 \leq j \leq 3),$$

and put

$$L = \text{lcm}(m, m+1, m+2, m+3), \quad d_j = \frac{L}{c_j}, \quad C_m = \prod_{j=0}^3 c_j^{e_j/2}, \quad T_m = \frac{4\pi^2}{L}.$$

The exponents e_j are primitive nonzero integers, and the raw product

$$E_m(t) = \prod_{j=0}^3 P(e^{-d_j t})^{e_j} \quad (7.8)$$

has the exact form

$$E_m(t) = C_m \mathcal{R}_m(e^{-T_m/t}), \quad \mathcal{R}_m(Q) = \prod_{j=0}^3 P(Q^{c_j})^{e_j} = 1 - \frac{m}{g} Q^m + \mathcal{O}(Q^{m+1}). \quad (7.9)$$

All three original balances vanish:

$$\sum e_j = 0, \quad \sum d_j e_j = 0, \quad \sum e_j/d_j = 0.$$

The nome multiplicity is exactly m . Four distinct factors are necessary for a nonzero exponent vector satisfying all three balances. For $m \geq 2$, the next coefficient is $3(m+1)/g$, so the effective nome support is primitive. For $m = 1$, primitiveness follows already from the nonzero linear coefficient.

Proof. The third finite difference of a polynomial of degree at most two vanishes. Applying it to $1, x, x^2$ gives

$$\sum_j \frac{e_j}{c_j} = 0, \quad \sum_j e_j = 0, \quad \sum_j c_j e_j = 0.$$

The gcd before dividing by g divides both m and $m+3$, and is therefore a divisor of $\gcd(m, 3)$. Conversely $\gcd(m, 3)$ divides all four entries, so the normalized vector is primitive.

The three balances in the original dilations follow from $d_j = L/c_j$:

$$\sum d_j e_j = L \sum e_j/c_j = 0, \quad \sum e_j/d_j = L^{-1} \sum c_j e_j = 0.$$

Apply [equation \(5.5\)](#) at $q = 1$. Its constant is

$$\prod_j d_j^{-e_j/2} = L^{-\sum e_j/2} \prod_j c_j^{e_j/2} = C_m,$$

and its dual powers are Q^{c_j} . Only the $j = 0$ factor contributes in degree m , giving $-m/g$. For $m \geq 2$, the next degree $m+1$ has only the $j = 1$ contribution, equal to $3(m+1)/g$. For $m = 1$, overlapping contributions change that coefficient to 5, but do not change the leading order.

To prove minimality, consider at most three distinct positive supports x_i . The balance matrix has rows $1, x_i, 1/x_i$. Multiplying its i -th column by x_i gives rows $x_i, x_i^2, 1$, a Vandermonde matrix up to row order. Its columns are independent for at most three distinct nonzero x_i . Hence its nullspace is zero. Our four nonzero factors attain the lower bound. $\square$

The same proof classifies all rational balanced vectors on four distinct positive rational supports c_0, c_1, c_2, c_3 : they are proportional to

$$e_j = \frac{c_j}{\prod_{\ell \neq j} (c_j - c_\ell)}. \quad (7.10)$$

Indeed Lagrange interpolation gives $\sum_j c_j^r / \prod_{\ell \neq j} (c_j - c_\ell) = 0$ for $r = 0, 1, 2$. These are exactly the three balances after multiplying by the appropriate power of c_j . The balance matrix has rank three, so the nullspace is one-dimensional. Clearing denominators and dividing by the gcd gives its primitive integer vector.

For $m = 3$, the construction yields dual data

$$(c_j) = (3, 4, 5, 6), \quad (e_j) = (1, -4, 5, -2), \quad (d_j) = (20, 15, 12, 10).$$

Here $L = 60$, $T_3 = \pi^2/15$, and $C_3 = 25\sqrt{15}/96$. The original four raw products have a constant perturbative expansion and a cubic nome defect.

For every $m \geq 2$, writing

$$y = \frac{1 - E_m/C_m}{m/g}, \quad s^m = y,$$

gives

$$Q = s + \frac{3(m+1)}{m^2} s^2 + \mathcal{O}(s^3).$$

The complete inverse and its certified tail are [equations \(6.4\)](#) and [\(6.8\)](#). The construction therefore supplies a family of explicit q -analog critical points of every order, together with convergent inverse expansions.

7.4 A low-level raw product

A compact additional example is

$$E(t) = \frac{P(e^{-2t})^5 P(e^{-6t})}{P(e^{-t}) P(e^{-3t})^5}.$$

The exponent vector on 1, 2, 3, 6 is $(-1, 5, -5, 1)$; its three balances are zero. Hence

$$E(t) = \frac{9}{8} \frac{P(Q) P(Q^3)^5}{P(Q^2)^5 P(Q^6)}, \quad Q = e^{-2\pi^2/(3t)}.$$

The normalized dual product is $1 - Q + 4Q^2 - 10Q^3 + 20Q^4 - 39Q^5 + \dots$. It gives a simple nome inverse and a logarithmic t -inverse. This example needs no eta prefactor and makes the distinction between raw and normalized observables concrete.

8 q -gamma at rational special values

The q -gamma function is defined, for $0 < q < 1$, by

$$\Gamma_q(x) = (1-q)^{1-x} \frac{(q; q)_\infty}{(q^x; q)_\infty} \quad [10, \text{Equation 5.18.4}].$$

For the complex sector used here, set $q = e^{-t}$, $q^x = e^{-tx}$, and continue the powers from positive t . For fixed $0 < x < 1$, all product factors remain nonzero when $\Re t > 0$. The explicit local branches near $t = 0$ suffice for the inverse results.

8.1 An exact modular completion for rational arguments

Theorem 8.1 (Rational-argument multiplication product). *Let $k \geq 2$ be an integer, and define*

$$\mathcal{G}_k(t) = \prod_{r=1}^{k-1} \Gamma_{e^{-t}}(r/k).$$

Then, for $\Re t > 0$, with the specified branches,

$$\mathcal{G}_k(t) = C_k(t) \mathcal{A}_k(Q), \quad Q = e^{-4\pi^2/t}, \quad \mathcal{A}_k(Q) = \frac{P(Q)^k}{P(Q^k)}, \quad (8.1)$$

where

$$C_k(t) = \frac{(2\pi)^{(k-1)/2}}{\sqrt{k}} \left(\frac{1 - e^{-t}}{t} \right)^{(k-1)/2} \exp \left(\frac{k - 1/k}{24} t \right). \quad (8.2)$$

The factor C_k is holomorphic and nonzero near $t = 0$, and $\mathcal{A}_k(Q) = 1 - kQ + \mathcal{O}(Q^2)$.

Proof. Splitting positive integers into their residue classes modulo k gives

$$\prod_{r=1}^{k-1} (q^{r/k}; q)_\infty = \frac{P(q^{1/k})}{P(q)}.$$

The powers of $1 - q$ add to $(k - 1)/2$. Consequently

$$\mathcal{G}_k(t) = (1 - e^{-t})^{(k-1)/2} \frac{P(e^{-t})^k}{P(e^{-t/k})}.$$

Apply the $q = 1$ modular formula separately to numerator and denominator. Their essential exponentials cancel exactly. The power constants give $(2\pi/t)^{(k-1)/2}/\sqrt{k}$; the regular exponential is $\exp((k - 1/k)t/24)$. The dual nome of the denominator is Q^k . This proves [equations \(8.1\)](#) and [\(8.2\)](#). The function $(1 - e^{-t})/t$ has removable value 1 at the origin, giving the stated analytic root. Finally, $P(Q) = 1 - Q + \mathcal{O}(Q^2)$, while $P(Q^k) = 1 + \mathcal{O}(Q^k)$. $\square$

Thus the *normalized observable*

$$\hat{\mathcal{G}}_k(t) = \mathcal{G}_k(t)/C_k(t)$$

has a flat cusp value 1. Setting $y = (1 - \hat{\mathcal{G}}_k)/k$, the analytic nome inverse starts $Q = y + \dots$, and then $t = -4\pi^2/\log Q$. This is an exact completion of the entire perturbative $q \rightarrow 1$ expansion for these multiplication products.

Every coefficient is arithmetic:

$$[Q^n] \log \mathcal{A}_k(Q) = -k\sigma_{-1}(n) + \mathbf{1}_{k|n} \sigma_{-1}(n/k). \quad (8.3)$$

Combining this with [equations \(5.8\)](#) and [\(6.2\)](#) computes the inverse to arbitrary order by finite rational arithmetic. For example,

$$\mathcal{A}_3(Q) = 1 - 3Q + 6Q^3 - 3Q^4 - 6Q^7 + 6Q^9 + \dots$$

8.2 The half-argument theta completion and a sparse inverse

For $k = 2$, $\mathcal{G}_2(t) = \Gamma_{e^{-t}}(1/2)$, and

$$\Gamma_{e^{-t}}(1/2) = \sqrt{\pi} \left(\frac{1 - e^{-t}}{t} \right)^{1/2} e^{t/16} \frac{P(Q)^2}{P(Q^2)}, \quad Q = e^{-4\pi^2/t}. \quad (8.4)$$

The Jacobi product gives

$$\frac{P(Q)^2}{P(Q^2)} = \vartheta_4(0, Q) = 1 + 2 \sum_{n \geq 1} (-1)^n Q^{n^2} = 1 - 2Q + 2Q^4 - 2Q^9 + 2Q^{16} - \dots \quad (8.5)$$

Here $\vartheta_4(0, Q)$ is the theta function with nome Q ; the product convention agrees with [\[11, Equation 20.5.4\]](#). This identity concerns a convergent Q -series, so its lacunarity is exact.

Let

$$y = \frac{1 - \hat{\mathcal{G}}_2(t)}{2} = Q - Q^4 + Q^9 - Q^{16} + \dots$$

Lagrange inversion yields

$$Q(y) = y + y^4 + 4y^7 - y^9 + 22y^{10} - 13y^{12} + 140y^{13} + \mathcal{O}(y^{14}), \quad (8.6)$$

$$\ell(y) = \log(Q(y)/y) = y^3 + \frac{7}{2}y^6 - y^8 + \frac{55}{3}y^9 - 12y^{11} + \frac{455}{4}y^{12} + \mathcal{O}(y^{13}). \quad (8.7)$$

Consequently the principal positive inverse is

$$t = \frac{4\pi^2}{\log(1/y) - \ell(y)}, \quad q = e^{-t}, \quad (8.8)$$

and all other small-nome branches are obtained by the coherent logarithm lifts in [equation \(6.4\)](#) with $\nu = 1$.

The sparse inverse has a simple support explanation. Writing

$$y = QH(Q), \quad H(Q) = 1 - Q^3 + Q^8 - Q^{15} + Q^{24} - \dots,$$

the support of $H - 1$ is $\{n^2 - 1 : n \geq 2\}$. Its additive monoid is exactly

$$\langle 3, 8 \rangle = \{3a + 8b : a, b \in \mathbb{N}\}.$$

The elements 3, 8 occur. If $3 \nmid n$, then $n^2 - 1$ is a multiple of 3; if $3 \mid n$, then $n^2 - 1 = 8 + (n^2 - 9)$, also in the monoid. It follows from [equation \(6.3\)](#) that every exponent of ℓ belongs to $\langle 3, 8 \rangle$. The absent powers are therefore structural, rather than accidental numerical zeros.

8.3 Reverting the original q -gamma value

The normalized flat observable and the raw q -gamma value have different inverses. For the latter set

$$\begin{aligned} w &= \log(\Gamma_{e^{-t}}(1/2)/\sqrt{\pi}), \\ G(t) &= \frac{1}{2} \log\left(\frac{1 - e^{-t}}{t}\right) + \frac{t}{16}, \quad L(Q) = \log \vartheta_4(0, Q). \end{aligned} \quad (8.9)$$

[Equation \(8.4\)](#) is exactly

$$w = G(t) + L(e^{-4\pi^2/t}).$$

The analytic background begins

$$G(t) = -\frac{3}{16}t + \frac{1}{48}t^2 - \frac{1}{5760}t^4 + \frac{1}{362880}t^6 + \dots, \quad G'(0) = -\frac{3}{16}. \quad (8.10)$$

Hence it has an ordinary local inverse

$$t_0 = G^{-1}(w) = -\frac{16}{3}w + \frac{256}{81}w^2 + \mathcal{O}(w^3).$$

Also $L(Q) = -2Q - 2Q^2 + \dots$. Applying [theorem 3.2](#) with $S = 4\pi^2$ gives

$$t = t_0 + \frac{2}{G'(t_0)}Q_0 + \sum_{n \geq 2} d_n(t_0)Q_0^n, \quad Q_0 = e^{-4\pi^2/t_0}. \quad (8.11)$$

The first correction is asymptotic to $-32Q_0/3$. For every $n \geq 1$,

$$d_n(t_0) \sim \frac{(-1)^n n^{n-1}}{n!} (4\pi^2)^{n-1} \left(\frac{32}{3}\right)^n t_0^{2-2n}. \quad (8.12)$$

The exact convergent form is

$$t = t_0 + t_0^2 U\left(t_0, \frac{e^{-4\pi^2/t_0}}{t_0^2}\right),$$

with the uniform error bound [equation \(3.11\)](#).

Thus the raw function has a nonzero sectorial derivative $-3\sqrt{\pi}/16$ at its limiting value. Its algebraic inverse is well defined as an asymptotic core, and the flat correction supplies the additional terms. The raw function does not become holomorphic across $t = 0$: the nonconstant flat factor in [equation \(8.4\)](#) still obstructs such an extension. The ordinary derivative and the essential boundary singularity are compatible.

For a direct inversion of the raw value rather than its logarithm, write $G_{\text{raw}}(t) = \sqrt{\pi}e^{G(t)}$. The first displacement is

$$\frac{2G_{\text{raw}}(t_0)}{G'_{\text{raw}}(t_0)}Q_0 = \frac{2}{G'(t_0)}Q_0,$$

so both descriptions agree.

The exponential action can also be expressed directly in the output deficit. Let

$$z = 1 - \frac{\Gamma_{e^{-t}}(1/2)}{\sqrt{\pi}}, \quad 1 - e^{G(t_0(z))} = z.$$

Ordinary analytic reversion of this background gives

$$t_0(z) = \frac{16}{3}z + \frac{472}{81}z^2 + \mathcal{O}(z^3), \quad \frac{1}{t_0(z)} = \frac{3}{16z} - \frac{59}{288} + \mathcal{O}(z).$$

Substitution into the first term of [equation \(8.11\)](#) therefore proves the explicit beyond-all-orders correction

$$t - t_0(z) \sim -\frac{32}{3}e^{59\pi^2/72}e^{-3\pi^2/(4z)} \quad (z \downarrow 0). \quad (8.13)$$

The transformed action $3\pi^2/4$ and its nontrivial constant prefactor both depend on retaining the correct analytic core.

The uniformity region is a strict decay sector for t_0 ; a path on which $\Re(4\pi^2/t_0)$ ceases to grow requires a different description.

8.4 Which notion of a critical point is involved?

There are several distinct questions involving q -gamma: varying its argument x at a fixed base, varying q toward a root of unity, removing a regular background, or letting both variables move. The special values above concern the base limit $q \rightarrow 1$ at fixed rational arguments. The ordinary minimum of $x \mapsto \Gamma_q(x)$, its motion with q , and a joint critical double scaling are different problems. The repository already contains results on that minimum and the ordinary base expansion [16]. Our theorem supplies exact modular sectors for specific rational multiplication products and a justified reversion of those sectors. It does not identify every critical point of a general q -special function.

9 Computation, certification, and reproducibility

The proofs establish convergence and branch validity. Computation has a separate role: detecting arithmetic, phase, and branch mistakes in the displayed examples. The archive contains exact rational coefficient generators and high precision checks; their numerical outputs are not used as substitutes for the analytic arguments.

9.1 A terminating coefficient calculation

For a prescribed degree N , the coefficient of Q^n in a finite phase Euler product is computed by equations (5.7) and (5.8). For unphased integer products it can also be computed independently by multiplying the finitely many relevant factors $(1 - Q^{dj})^{r_d}$, since factors with $dj > N$ do not affect that truncation. Negative powers use the ordinary generalized binomial coefficients.

Once the first surviving order ν is known, form

$$H(Q) = \frac{R(Q) - 1}{c_\nu Q^\nu}.$$

Computing Ψ through degree N requires H only through degree $N - 1$. Computing ℓ through degree N requires H through degree N . Each rational-power coefficient is obtained from

$$(1 + Z)^\alpha = \sum_{j=0}^N \binom{\alpha}{j} Z^j \pmod{Q^{N+1}},$$

with a finite sum. Exact composition then verifies $R(\Psi(u)) = 1 + c_\nu u^\nu$ through the available order.

In the supplied program, both Euler-product methods agree, the inverse compositions agree exactly, and the arbitrary-order four-factor balances and primitive vectors are checked for $m = 1, \dots, 7$. These checks target independent representations of the formulas.

9.2 A concrete certified disk for the eta example

The hypotheses of theorem 6.2 can be verified with rational inequalities, not merely numerical sampling. For equation (7.4), put $H(Q) = (1 - R_2(Q))/(3Q^2)$, and choose $r = 1/20$. On $|Q| = r$, let

$$\begin{aligned} E_r &= 3 \frac{r^4(2 - r^2)}{(1 - r^2)^2} + 4 \frac{r^3}{(1 - r^3)^2} + \frac{r^6}{(1 - r^6)^2}, \\ T_r &= 3r^2 + E_r. \end{aligned}$$

The first quantity bounds $|\log R_2(Q) + 3Q^2|$: in the $P(Q^2)^3$ term use $\sigma_{-1}(n) \leq n$ after removing degree two. The other two factors use equation (5.9). Consequently $|\log R_2| \leq T_r < 1$, and

$$|R_2 - 1 - \log R_2| \leq \frac{1}{2} T_r^2 e^{T_r} \leq \frac{T_r^2}{2(1 - T_r)}.$$

The maximum-modulus principle applied to the removable analytic function $H - 1$ gives

$$\sup_{|Q| \leq r} |H(Q) - 1| \leq \frac{E_r}{3r^2} + \frac{T_r^2}{6r^2(1 - T_r)} < \frac{1}{2}. \quad (9.1)$$

The final comparison is an exact rational inequality; its left side bound is approximately 0.0760461945891084, and the supplied rational-arithmetic calculation proves that it is less than $760463/10^7 < 1/2$. Thus $R_0 = 1/(20\sqrt{2})$ and $M = (\log 2)/2$ are explicit valid constants in theorem 6.2. The numerical cutoff is conservative, but it is completely reviewable.

For numerical evaluation of a dual factor, truncating $P(z)$ after J factors has logarithmic tail bounded by

$$\left| \sum_{n>J} \log(1 - z^n) \right| \leq \frac{|z|^{J+1}}{(1 - |z|)(1 - |z|^{J+1})}. \quad (9.2)$$

Indeed expand each logarithm, sum the geometric tail in n , and bound $1 - |z|^\ell$ below by $1 - |z|$. A sum of these bounds, weighted by the absolute factor exponents, controls a quotient's logarithm. To convert a logarithmic error ε to relative multiplicative error, use $|e^\varepsilon - 1| \leq |\varepsilon|e^{|\varepsilon|}$.

9.3 Numerical examples and logarithm winding

The principal eta inverse [equation \(7.5\)](#) was evaluated at 360 decimal digits. The following table uses the polynomial for $Q(s)$ through degree 10, followed by the coherent logarithm $t = (2\pi^2/3)/(-\log Q)$. It therefore reports a Q -polynomial truncation, rather than the ℓ -polynomial truncation in [equation \(6.8\)](#).

Original parameter t	Observed absolute inverse error
0.8	3.33249×10^{-35}
0.5	4.82563×10^{-57}
0.3	1.37812×10^{-95}
$0.5 + 0.15i$	2.75344×10^{-52}

The complex example requires logarithmic winding index -1 when its nome is first represented with the principal logarithm. Resetting that logarithm would return a different inverse sheet. At the same output as $t = 0.8$, the other small nome root gives, on one logarithm lift,

$$t_- \approx 0.6981170734058837 + 0.2666494981906634i.$$

Direct evaluation of the original eta quotient agrees for these two parameters to the working numerical precision. The existence of both branches follows from [theorem 6.1](#); the computation verifies the sheet bookkeeping in a concrete case.

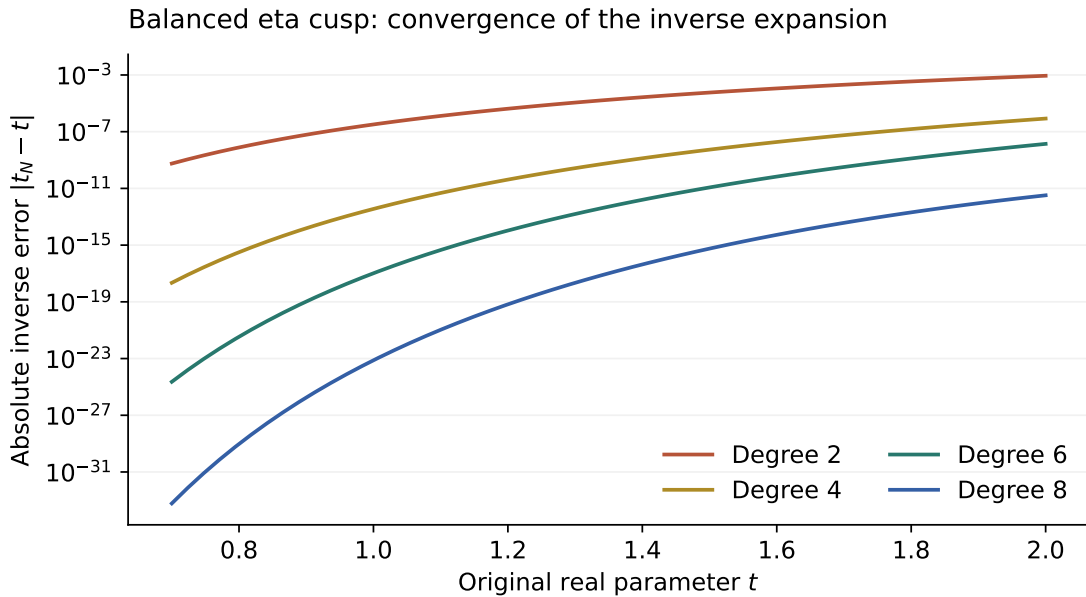


Figure 1: Observed error after truncating the nome inverse $Q(s)$ at degrees 2, 4, 6, 8, then taking a coherent logarithm. Each curve uses direct high precision evaluation of the eta example. Moving toward the cusp makes the exponential coordinate smaller. The curves are numerical illustrations; the analytic tail bounds are [equations \(6.8\)](#) and [\(9.2\)](#).

The archive also records the rational q -gamma product checks for $k = 2, 3, 4$, the four-factor family at real and complex arguments, explicit rational-cusp phase checks, and paired-term Stokes contour checks. The latter compare the two directional integrals with their residue sum; the counterexample itself is established by its contour proof, independently of those calculations.

The supplementary phase program checks 24 one-factor and 24 general-product identities, with both real and complex parameters, at 80 decimal digits. The largest scaled residual is less than 1.3×10^{-79} . It also checks Dedekind reciprocity with exact rational arithmetic and records the rational certificate in [equation \(9.1\)](#). The Stokes program independently integrates four paired summands along the two rays; each difference agrees with its residue sum to absolute error less than 4×10^{-82} . These are reproducible checks at specified working precision; the numerical residuals are not interval-arithmetic certificates.

9.4 An implementation procedure

A reliable implementation can follow this sequence.

1. Record the input function, variable being inverted, cusp, target normalization, sector, and logarithm lift.
2. For a finite eta or Euler quotient, compute $g_d, \rho_d, A_{h,k}, B, D$ exactly. Select the appropriate core regime.
3. Combine coincident action coefficients in a cyclotomic field before identifying the first surviving term.
4. Use the exact-core theorem, the finite-endpoint scale, or the flat-cusp inverse according to that regime. Near a zero of the core derivative, change the local chart.
5. Generate coefficients to the desired order, verify composition algebraically, and compute a disk or tail bound.
6. Evaluate products through the dual nome, preserving logarithm lifts. If the target comes from a Borel sum, carry the contour's residue correction into the equation.

This procedure makes two common failure modes explicit. Deleting a flat modular factor may remove the entire inverse problem after balancing. Ignoring a Stokes correction may invert a different sectorial function while preserving the same formal power series.

10 Further research questions and precise remaining boundaries

The article supplies exact results for substantial classes, but a general complex transseries program has additional problems. The following questions separate those problems by the missing mathematical ingredient.

Research question 10.1 (Uniform parameter-dependent resurgent inversion). *Starting with [equation \(4.7\)](#), prove the full family of resurgent seminorm bounds needed for implicit inversion in a , uniformly on compact subsets of $|a| < 1$, with specified contours and target jumps. Identify a natural function space in which both the contour correction and the inverse depend holomorphically on the parameters.*

The meromorphic Borel transform and the exact equation [equation \(4.22\)](#) are available. A pointwise nonzero derivative does not by itself supply uniform inverse radii or all of the needed functional analytic estimates. This is the part of the repository's implicit conjecture that has not been replaced here by an unconditional general theorem.

Research question 10.2 (Coalescing fixed-argument and modular limits). *Obtain a uniform inverse theory when $a = a(t)$ approaches the unit circle, for example $a(t) = e^{-\alpha t}$, while $q = \zeta e^{-t} \rightarrow \zeta$. How do the Borel pole arrays reorganize into the modular completions when α is rational, and what replaces those completions for general complex α ?*

The fixed compact- a estimates deteriorate as $|a| \rightarrow 1$. The exact eta and rational q -gamma identities avoid that deterioration in special cases but do not prove a uniform transition theorem for arbitrary α . A useful first goal is a two-parameter sector with an explicit matching estimate on an overlap.

Research question 10.3 (Moving cusps with growing denominator). *For fixed dilations and rational cusps h/k depending on t , determine uniform reversion and error bounds across the transition $k^2 t \asymp 1$, including the phase arithmetic.*

The exact normal form remains valid. What fails is the conclusion $Q \rightarrow 0$. A transition approximation must retain a non-small dual product, and cannot follow from a finite number of its coefficients. Uniform control in h and k is a separate issue from the fixed-cusp formula.

Research question 10.4 (Effective cancellation certificates). *Given a finite phase Euler product, find practical bounds on the first nonzero coefficient of its logarithm, or prove that it is identically zero. Can modular-form bounds and arithmetic coefficient recurrences be combined to outperform direct expansion, including products with nontrivial eta multipliers?*

The constructions here have visible first coefficients. For arbitrary phase products, cancellation can be extensive or complete. An effective certificate should also identify the primitive action lattice, so that a reported ramification order is intrinsic to the chosen minimal nome.

Research question 10.5 (Optimal level and height for a prescribed ramification). *Among all balanced products realizing order m , minimize the common level, the largest original dilation, or the sum of the absolute exponents. Determine the tradeoff between those quantities under two versus three balance constraints.*

The factor counts three and four are sharp. They do not optimize the level of the four-factor construction, which uses an lcm of four consecutive integers. The one-dimensional classification [equation \(7.10\)](#) supplies an exact starting point for the optimization.

Research question 10.6 (Global continuation and coefficient asymptotics). *For a specific balanced quotient, locate the first singularities of $\Psi(u)$, including critical values of the dual product and images of other cusps. Determine the exact convergence radius and the large-order asymptotics of its inverse coefficients.*

The local inverse theorem gives a positive radius, not its optimal value. Modular coordinates may permit a global analysis in low-level examples. Such an analysis requires a verified description of the relevant modular curve or covering and should not be inferred from a few inverse coefficients.

Research question 10.7 (Unfolding a flat critical point). *Let a parameter break one or more balance equations, so that*

$$F_\varepsilon(t) = C t^{-b(\varepsilon)} e^{-a(\varepsilon)/t + c(\varepsilon)t} R_\varepsilon(e^{-S/t}).$$

Construct uniform inverse charts as the algebraic core and the flat defect become comparable.

Theorems for each separate regime do not automatically remain uniform when $a, b, c \rightarrow 0$. Natural transition variables compare a/t , $b \log t$, and ct with the first surviving Q -action. This is a singular perturbation problem rather than an ordinary fixed-parameter application of the inverse-function theorem.

Research question 10.8 (Finite moving critical points and higher multiplicity). *Extend the present exact examples to a parameter-dependent critical point of multiplicity m , with explicit moving discriminants, sheet radii, and crossover bounds.*

The generic local Puiseux theorem is already available in the repository. A perturbed critical point may split. For instance $u^2 + \varepsilon u = y$ has its critical value at $-\varepsilon^2/4$, so an expansion about the undeformed critical value is not uniform. The arbitrary- m cusp constructions do not solve that full moving-discriminant problem.

Research question 10.9 (Infinite and accumulating complex actions). *Replace the finite analytic action set by a countable one, possibly with accumulation in exponential height. Find conditions under which coefficient grouping, analytic substitution, and inverse summation remain compatible.*

The finite decay-cone argument bounds the number of indices at each height. It does not extend merely by writing an infinite list of actions. A viable theorem needs quantitative summability of grouped coefficients and control of the inverse iteration.

Research question 10.10 (More q -special functions and joint criticality). *Determine which rational combinations of q -beta, q -gamma, q -Barnes functions, and Gaussian coefficients admit exact modular completions with enough balance to create flat critical values. Study joint argument/base critical limits.*

The product identities in [section 8](#) provide a testable model: first isolate an exact analytic background, then identify the surviving modular coordinate and apply the appropriate inverse theorem. Existing questions on optimal q -multinomial truncation, pole pinching, and the motion of the q -gamma minimum remain distinct [\[18, 16\]](#).

Research question 10.11 (Formal verification of the analytic boundary). *Formalize the finite arithmetic recurrences and balance constructions, then connect them to a verified analytic inverse theorem and explicit complex product bounds.*

The rational coefficient generator is a suitable first target. A full proof also needs complex branch data, normal convergence, Rouché or contraction certificates, and the modular identity. No theorem in this article is claimed to have been checked by a proof-assistant kernel. The included programs check exact finite algebra and numerical examples.

10.1 What is settled here

The directional-equality assertion in the named repository conjecture has an explicit counterexample, while its fixed-argument Gevrey and Borel-continuation content has a direct, literature-consistent proof. The complete-cancellation question has explicit arbitrary-order examples with sharp factor counts. The inverse charts have all coefficients, all small-nome logarithmic sheets, and quantitative tails. The rational q -gamma applications distinguish the flat normalized inverse from the regular-core inverse of the original observable.

These conclusions are compatible with the limits stated throughout: fixed sectors and branches are part of the data; a fixed-cusp formula is not a growing-denominator theorem; local convergence is not global continuation; and a finite analytic action model is not the entire field of complex transseries.

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